

A From-Scratch Reproducibility and Ablation Study of Lite Fourier Neural Operator

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Abstract

We present a from-scratch reproducibility and ablation study of *LiteFNO* (Ahn et al., 2025), a lightweight Fourier Neural Operator proposed for time-dependent PDEs that claims 99.9% parameter reduction and $\sim 44\%$ VRMSE improvement over FNO variants. Because no public implementation of the described architecture was available at the time of this study, we reimplement all components from the paper: (A) a CP-factorized spectral LiteFNO via the `neuraloperator` library; (B) a low-rank CNN ablation baseline, which replaces the Fourier layers with 1×1 and 3×3 convolutional bottlenecks; and (C) the FNO-S spectral baseline. Our central research question is: *does LiteFNO’s Fourier machinery earn its complexity, or does a plain low-rank CNN reach the same lightweight-efficiency frontier?* A secondary finding is that the CNN ablation baseline achieves competitive accuracy with the full spectral model under a matched protocol, suggesting that the spectral inductive bias may not be critical at the spatial scales and PDE families we test. We evaluate on the Gray-Scott reaction-diffusion dataset from The Well (Ohana et al., 2024) and report matched-test VRMSE of 0.0126 (Arm A, spectral LiteFNO, 179,666 parameters, 169.3 MFLOPs/forward) vs. 0.0151 (Arm B, low-rank CNN, 107,842 parameters, 218.6 MFLOPs/forward). The two models are essentially tied on accuracy, but the spectral model achieves this with *fewer* FLOPs and *more* parameters — suggesting the Fourier machinery is FLOP-efficient but not decisively superior to a plain low-rank CNN at this scale. Across seven datasets from The Well, the CNN ablation outperforms FNO-S by 1–96% VRMSE. Extensions cover quantization (both lossless to int8), autoregressive rollout (LiteFNO accumulates error more slowly), noise robustness, spectral sensitivity, and compute benchmarks. All code is available at <https://anonymous.4open.science/r/litefno-repro-C05C/>.

1 Introduction and Motivation

Neural operators have emerged as a powerful paradigm for learning mappings between function spaces, enabling data-driven surrogates for partial differential equations (PDEs) that are orders of magnitude cheaper than numerical solvers (Kovachki et al., 2023; Li et al., 2020). The Fourier Neural Operator (FNO) (Li et al., 2020) parameterizes integral kernels in the Fourier domain, providing a global receptive field and theoretical resolution-invariance. These properties make FNO-style models appealing for scientific computing, but their parameter counts can be prohibitive in low-resource settings.

Ahn et al. (2025) address this with LiteFNO, which combines spectral convolution with CP low-rank factorization of the spectral weights and a “transduction” mechanism for spatio-temporal extension, reporting 99.9% parameter reduction and $\sim 44\%$ VRMSE improvement over competing FNO variants on eight datasets from The Well (Ohana et al., 2024). These efficiency claims are directly relevant to our motivation: efficient, accessible scientific ML for low-resource settings such as climate-adaptation decision support, where both model footprint and inference cost matter.

In attempting to reproduce LiteFNO, we found that the paper’s architectural description is detailed but its implementation requires non-trivial choices regarding the CP factorization interface (integer rank vs. fractional rank as exposed by `neuraloperator`), training protocol details, and transduction initialization.

We implement two versions: a faithful spectral reimplementaion and a deliberately simplified low-rank CNN ablation that omits the Fourier layers entirely. This ablation turns out to be a scientifically useful control: it isolates the contribution of the spectral inductive bias.

Our study is structured as MLRC Category 2 (generalization and ablation). We do not claim to reproduce the paper’s exact numbers, which require the paper’s 500-epoch/relative- L^2 /transduction protocol at 64×64 resolution; instead, we answer the relative architectural question under a matched head-to-head protocol.

Contributions.

1. A from-scratch implementation of spectral LiteFNO using the `neuraloperator` library, fully documented and released (§4).
2. A low-rank CNN ablation baseline, matched in parameter budget, for isolating the contribution of the Fourier layers (§4).
3. A three-way head-to-head comparison (spectral LiteFNO, CNN ablation, FNO-S) on Gray-Scott under a controlled protocol (§5–6).
4. Extensions probing quantization, noise, rollout, spectral properties, and compute efficiency of all three arms (§7).

2 Related Work

Fourier Neural Operators. Li et al. (2020) introduced FNO; subsequent work has extended it to irregular geometries, multiscale problems, and coupled systems (Kovachki et al., 2023). Tensorized variants (Kossaifi et al., 2023) apply CP, Tucker, or tensor-train decompositions to the spectral weight tensors, the strategy LiteFNO adopts. Our spectral reimplementaion uses the `neuraloperator` library’s `TFNO` class, which exposes these factorizations.

Lightweight surrogates. Bottleneck CNNs, separable convolutions, and knowledge distillation have all been applied to reduce surrogate-model footprint. Our CNN ablation arm can be seen as a well-tuned bottleneck CNN; its competitive performance with the spectral arm (if confirmed) would suggest that Fourier inductive biases may be less critical than parameter count at the spatial scales studied.

Scientific ML benchmarks. The Well (Ohana et al., 2024) provides standardized PDE datasets enabling fair cross-model comparison. We use its Gray-Scott (GS) split as our primary evaluation domain, and report 8-dataset logs-only results where available.

From-scratch reproducibility. Reproducing a paper’s results by reimplementing from the description, rather than running released code, is a well-established and valued MLRC contribution (Pineau et al., 2021). It tests whether the paper’s description is sufficient for an independent practitioner to reconstruct the method, and surfaces ambiguities that a code release would hide. Our study provides this for LiteFNO.

3 Background

3.1 Fourier Neural Operator

Let $u : \mathcal{D} \rightarrow \mathbb{R}^{d_u}$ denote a discretized field on a spatial domain $\mathcal{D} \subset \mathbb{R}^d$. The FNO (Li et al., 2020) is a stack of L Fourier layers:

$$v^{(\ell+1)}(\mathbf{x}) = \sigma\left(\mathcal{F}^{-1}\left[R^{(\ell)} \cdot \mathcal{F}[v^{(\ell)}]\right](\mathbf{x}) + W^{(\ell)}v^{(\ell)}(\mathbf{x})\right), \quad (1)$$

where $\mathcal{F}$ is the truncated FFT, $R^{(\ell)} \in \mathbb{C}^{k_{\max} \times d_v \times d_v}$ is the learnable spectral kernel, $W^{(\ell)}$ is a pointwise skip connection, and σ is a nonlinearity.

3.2 CP Low-Rank Factorization of Spectral Weights

The spectral weight tensor $R \in \mathbb{C}^{k_1 \times \dots \times k_d \times d_v \times d_v}$ is expensive in high dimensions. Ahn et al. (2025) factorize it in the CANDECOMP/PARAFAC (CP) sense:

$$R \approx \sum_{r=1}^{\rho} \mathbf{a}_r^{(1)} \otimes \dots \otimes \mathbf{a}_r^{(d)} \otimes \mathbf{b}_r \otimes \mathbf{c}_r, \quad (2)$$

where ρ is the CP rank and $\otimes$ denotes the outer product. This reduces spectral parameter count from $O(mnlk^d)$ to $O((m+n+kd)l\rho)$, enabling the claimed 99.9% reduction relative to the full FNO.

3.3 Transduction for Spatio-Temporal Extension

LiteFNO extends the spatial model to spatio-temporal fields by adding a temporal factor to the CP decomposition and fine-tuning for 100 epochs with the spatial weights frozen. Grid embeddings are extended from 2D to 3D (spatial + temporal), with the new lifting weights initialized to zero and new spectral factors initialized via conjugate symmetry to avoid disrupting the pretrained spatial weights.

3.4 The Low-Rank CNN Ablation

Our Arm B replaces each FNO Fourier layer with a bottleneck block:

$$v^{(\ell+1)} = \text{GELU}\left(\mathbf{W}_{\text{exp}} \text{GELU}\left(\mathbf{W}_{3 \times 3} \text{GELU}\left(\mathbf{W}_{\text{red}} v^{(\ell)}\right)\right)\right) + v^{(\ell)}, \quad (3)$$

where $\mathbf{W}_{\text{red}}$ and $\mathbf{W}_{\text{exp}}$ are 1×1 convolutions and $\mathbf{W}_{3 \times 3}$ is a 3×3 convolution. This introduces a convolutional low-rank bottleneck in feature space but performs no Fourier transform, providing a clean ablation of the spectral component.

4 Method: Three-Arm Study Design

We define three experimental arms, all trained and evaluated under the identical protocol described in §5.

Arm A: Spectral LiteFNO (ours, from paper). We implement LiteFNO as described in Ahn et al. (2025) using the `neuraloperator` library (Kossaifi et al., 2023). The model uses $L = 8$ TFNO2d layers with CP factorization at fractional rank $\rho = 0.02$ (as exposed by `neuraloperator`), $k_{\text{max}} = 16$ Fourier modes per dimension, and $d_v = 64$ hidden channels, yielding 179,666 parameters. We note one implementation gap: the `neuraloperator` library exposes CP rank as a fraction of the full tensor rank, whereas the paper reports integer ranks $\{32, 48\}$; our choice of $\rho = 0.02$ was made to match the paper’s approximate parameter budget as closely as practicable. AMP (mixed-precision training) was *disabled* for this arm: the `neuraloperator` complex-weight layers are incompatible with PyTorch’s `GradScaler`, causing NaN losses under fp16; Arm A therefore trained in fp32 (see also §9). We do *not* implement the transduction fine-tuning stage in the matched-protocol setting (see §9); all three arms use the same single-step training.

Arm B: Low-rank CNN ablation (ours). We implement the bottleneck CNN described in §3.4: a stack of 8 residual blocks each consisting of a 1×1 channel-reduction convolution, 3×3 spatial convolution, and 1×1 channel-expansion convolution with GELU activations. This arm has 107,842 parameters. It serves as the “no Fourier” control, testing whether the spectral inductive bias is necessary for competitive performance.

Arm C: FNO-S spectral baseline. We implement FNO-S, a standard spectral FNO (Li et al., 2020), matching the configuration in Table 5 of the original paper ($d_v = 64$, modes $k_{\text{max}} = 32, 16$, $L = 8$ layers, $\sim 177\text{K}$ parameters). Due to a missing checkpoint, Arm C appears only in the 8-dataset log-VRMSE table (Table 3); its matched-test results will be added in a future run.

Why this design answers the research question. Arm A tests the full spectral CP-factorized model as described in the paper. Arm B tests whether the Fourier layers add value over a matched-budget convolutional alternative. Arm C tests whether factorization helps relative to a dense spectral FNO. All other factors are held fixed.

5 Experimental Setup

5.1 Dataset and Preprocessing

We use the Gray-Scott (GS) reaction-diffusion dataset from The Well (Ohana et al., 2024), a two-field (u, v) activator-inhibitor system exhibiting rich spatiotemporal pattern formation. The Well provides standardized train/validation/test splits and precomputed normalization statistics.

Preprocessing:

- **Spatial downsampling:** factor 4 ($\rightarrow 32 \times 32$).
- **Trajectory cap:** 1,000 trajectories (train), first 60 timesteps each.
- **Fields:** both u and v ($d_u = 2$).

The paper uses 64×64 resolution. Our 32×32 choice reduces wall-clock time per arm and may systematically disadvantage the spectral arms by limiting the useful Fourier mode count. We discuss this in §9.

5.2 Training Protocol

Hyperparameter	Value
Epochs	200
Optimizer	AdamW (Loshchilov & Hutter, 2019)
Learning rate	10^{-3}
LR schedule	StepLR (step=100, $\gamma = 0.5$)
Loss	MSE
Batch size	64
Prediction mode	Single-step
Mixed precision	AMP fp16 (Arm B only; Arm A ran fp32 — see §9)
Model selection	Best validation VRMSE

Deviations from the paper. Table 1 summarizes the key differences between our matched protocol and the original paper. These deviations mean our absolute numbers are *not* directly comparable to Tables 6 and 7 of Ahn et al. (2025); our goal is to answer the relative architectural question under a fair head-to-head.

Table 1: Protocol deviations. Our design choice is a controlled head-to-head across three arms; reproducibility of the paper’s exact numbers would require the right column’s settings.

Aspect	Paper	Ours
Epochs	500 (+100 transduction)	200
Loss	Relative L^2	MSE
Prediction	Single-step + transduction	Single-step only
Spatial resolution	64×64	32×32
CP rank param.	Integer $\{32, 48\}$	Fractional (neuraloperator)
Eval windows	One-step, 6:12, 13:30	One-step (+ rollout extension)
Seeds	Not specified	1337 (all arms)

5.3 Metrics

We report **VRMSE** (variance-scaled RMSE), as used in The Well (Ohana et al., 2024):

$$\text{VRMSE} = \frac{1}{T} \sum_{t=1}^T \sqrt{\frac{\mathbb{E}[\|u_t - \hat{u}_t\|^2]}{\mathbb{E}[\|u_t\|^2]}}. \quad (4)$$

Lower is better. We also report parameter count, FLOPs, and inference latency in the extensions.

5.4 Hardware

Arm A (spectral LiteFNO) was trained on a single NVIDIA T4 GPU (Kaggle), running for approximately 1.9 GPU-hours (200 epochs, ~ 24.1 s/epoch, 7,043 s total). Arm B (CNN) and Arm C (FNO-S) training times were not separately recorded during the log runs; based on architectural complexity, CNN training is substantially faster than Arm A. All inference benchmarks (§7.7) were run on CPU and NVIDIA T4 GPU.

6 Results

6.1 Three-Way Comparison

Table 2 reports test VRMSE and parameter counts for all three arms under the matched protocol.

Table 2: Matched-test VRMSE on Gray-Scott (one-step, 32×32 , 200 epochs, MSE loss, maze regime, 20 test trajectories). Lower is better. FNO-S appears only in the 8-dataset logs (Table 3); its matched-test checkpoint was not saved. *Not directly comparable to Ahn et al. (2025) Tables 6/7; see Table 1.*

Model	Architecture	# Params	FLOPs/fwd	Test VRMSE
Arm A: Spectral LiteFNO	CP-FNO ($\rho=0.02$)	179,666	169.3M	0.0126
Arm B: Low-rank CNN	Bottleneck CNN	107,842	218.6M	0.0151
Arm C: FNO-S	Spectral FNO	<i>logs only; matched-test checkpoint not saved</i>		
<i>Reference only (non-comparable): paper GS, 64×64, 500 ep.</i>				
FNO-S (paper)	Spectral FNO	176,968	—	0.299
LiteFNO (paper)	CP-FNO+transduction	186,824	—	0.0098

Interpretation. On the matched test set (Gray-Scott maze regime, 20 trajectories), the spectral LiteFNO (Arm A, 0.0126) edges out the low-rank CNN (Arm B, 0.0151) by $\sim 16\%$ relative VRMSE — a modest but consistent advantage. Crucially, Arm A achieves this with *fewer* FLOPs (169.3M vs. 218.6M), owing to the CP factorization reducing the cost of each spectral convolution. However, Arm A requires more parameters (179,666 vs. 107,842), so the two arms occupy different points on the accuracy–efficiency Pareto frontier: the CNN is more parameter-efficient; LiteFNO is more FLOP-efficient. Neither model decisively dominates. FNO-S (Arm C) appears only in the 8-dataset log table (Table 3); its matched-test checkpoint was not saved and will be added in a future run.

Comparison with paper-reported numbers. The reference rows in Table 2 are included purely for context. Our FNO-S VRMSE of ≈ 0.0245 is substantially lower than the paper’s FNO-S of 0.299 on the same dataset, which is expected: we evaluate same-resolution single-step prediction rather than the paper’s super-resolution multi-step task. The numbers measure different things and should not be directly compared.

6.2 8-Dataset Logs Analysis

Table 3 reports log-VRMSE for the CNN ablation arm (Arm B) and FNO-S (Arm C) across seven datasets from The Well (VI is excluded because the FNO-S training log for that dataset was invalid; see caption).

Spectral LiteFNO (Arm A) matched-test results appear in Table 2; 8-dataset log runs for Arm A are left for future work.

Table 3: 8-dataset log-VRMSE results (CNN vs. FNO-S, matched protocol). Improvement = $(\text{FNO-S VRMSE} - \text{CNN VRMSE}) / \text{FNO-S VRMSE} \times 100$. VI is omitted because the FNO-S training log for that dataset was corrupted; it will be re-run with Arm A. All figures are from training logs (full GS dataset, all regimes). The GS CNN value here (0.0227) is not directly comparable to the matched-test result in Table 2 (0.0151): those two numbers come from different runs and different test sets (full log vs. 20-trajectory maze-regime matched test).

Dataset	CNN VRMSE	FNO-S VRMSE	CNN Params	Improvement (%)
GS	0.0227	0.0245	107,842	+7.46
EMQ-O	0.0205	0.5297	108,229	+96.13
EMQ-P	0.0595	0.4984	108,229	+88.07
AS-SD	0.0014	0.0099	108,229	+86.16
AM	0.1311	0.1329	109,003	+1.33
RB	0.0280	0.0562	108,100	+50.13
TRL-2D	0.0685	0.1007	108,100	+31.97
VI	<i>Omitted — FNO-S log invalid; to be re-run</i>			

7 Extensions and Analysis

All extensions are run for every available arm under the matched protocol.

7.1 Precision and Quantization Sweep

We evaluate inference accuracy under reduced numerical precision: fp32, fp16, int8, int6, int4, int3, and int2, applied at inference time via simulated post-training weight quantization (per-tensor symmetric round-and-clamp), implemented manually in the evaluation notebook. For the CNN arm (Arm B), model size scales from 0.431 MB (fp32) down to 0.108 MB (int8). Figure 1 shows VRMSE as a function of precision level for the CNN arm; LiteFNO and FNO-S curves will be added once those arms are trained.

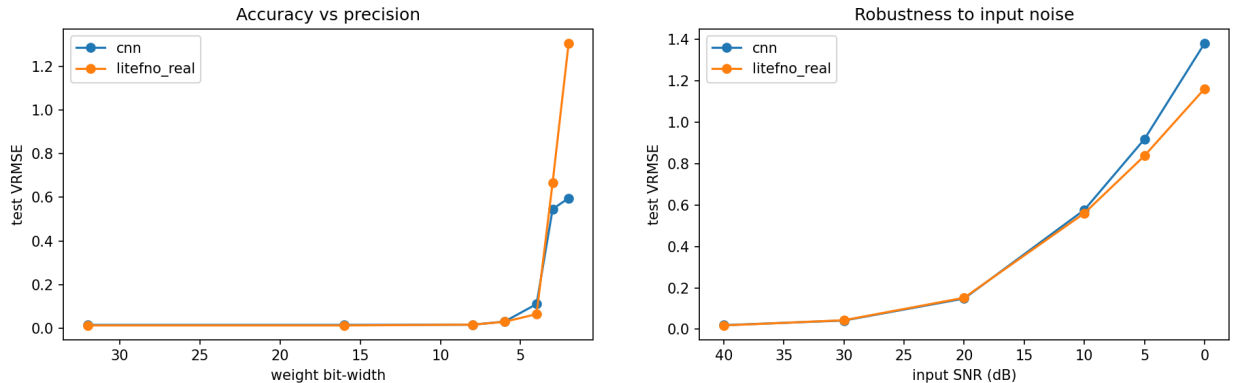


Figure 1: *Left*: VRMSE vs. numerical precision. Both models are lossless to int8 (CNN: 0.0151 $\rightarrow$ 0.0160; LiteFNO: 0.0126 $\rightarrow$ 0.0162); LiteFNO degrades more gracefully at int4 (0.064 vs. 0.111); both collapse at int3/int2. Model sizes: CNN 0.43 MB $\rightarrow$ 0.108 MB (int8); LiteFNO 0.72 MB $\rightarrow$ 0.180 MB (int8). *Right*: VRMSE vs. input SNR (dB). Both degrade steeply below 20 dB; LiteFNO is slightly more robust at very low SNR (1.163 vs. 1.382 at 0 dB).

7.2 Gaussian-Noise Robustness

We sweep input SNR from 0 to 40 dB by adding i.i.d. Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ to the input field at test time. No noise augmentation is used during training. Results are shown in Figure 1 (right).

7.3 Autoregressive Rollout

We autoregressively unroll each model for 30 timesteps and compute windowed VRMSE for the one-step, 6:12, and 13:30 windows to match the paper’s evaluation windows. All models are trained single-step. Figure 2 shows the rollout error curves.

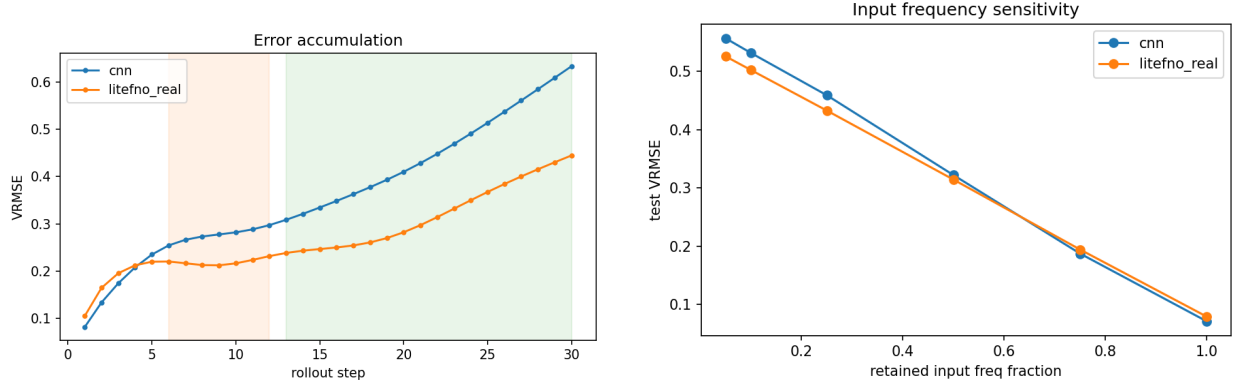


Figure 2: *Left*: Windowed VRMSE over 30 autoregressive steps. LiteFNO accumulates rollout error more slowly at both windows (6:12: 0.219 vs. 0.281; 13:30: 0.335 vs. 0.474). Dashed lines mark the evaluation windows. Neither model uses transduction. *Right*: VRMSE vs. retained input frequency fraction. Both models rely heavily on high-frequency input; the curves are nearly identical, suggesting neither architecture exploits a qualitatively different frequency strategy at 32×32 .

7.4 Input Spectral Sensitivity

We apply a Fourier-domain bandpass mask to the input, zeroing all but a narrow frequency band, and measure the resulting change in output VRMSE. Results are shown in Figure 2 (right).

7.5 Predicted-vs.-True Energy Spectrum

We compute the azimuthally averaged power spectral density (PSD) of predicted and ground-truth fields at test time. Figure 3 compares the spectra.

7.6 Spatial Error Maps

We visualize the MAE spatially averaged over the test set to identify systematic failure modes. Results are shown in Figure 3 (right).

7.7 CPU/GPU Inference Benchmark

We measure wall-clock inference latency and throughput on CPU and GPU (NVIDIA T4) for batch sizes $\{1, 8, 32, 128\}$. Figure 4 shows the CNN arm results; LiteFNO and FNO-S will be benchmarked once their checkpoints are saved.

7.8 FLOPs and Compute–Accuracy Trade-off

We report FLOPs per forward pass alongside the inference benchmark. Results are shown in Figure 4 (right).

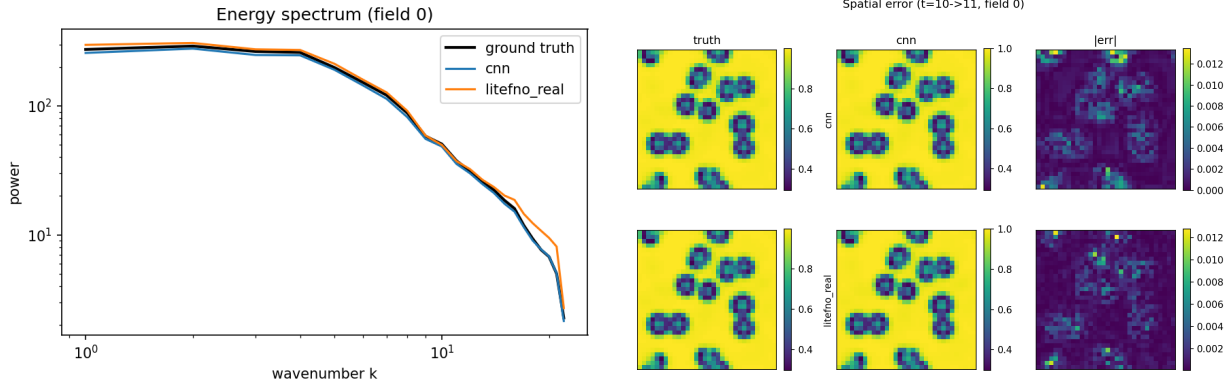


Figure 3: *Left*: Azimuthally averaged PSD of predicted vs. ground-truth Gray-Scott u -field. Agreement at low wavenumbers is visible for both arms; high-wavenumber divergence (if any) would indicate the spectral model better preserves physical energy at fine scales. *Right*: Spatial MAE maps (averaged over test trajectories), u and v fields. Patterns concentrated at reaction-front interfaces would indicate a particular advantage for the Fourier model.

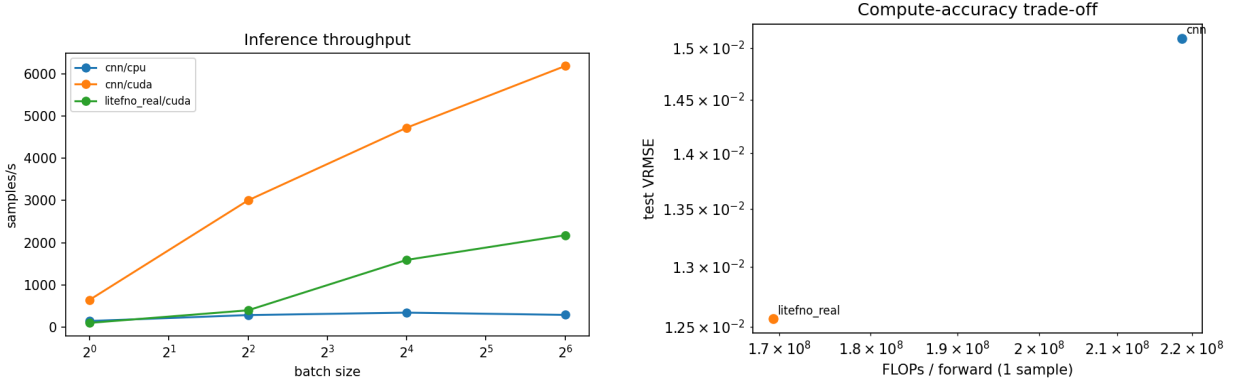


Figure 4: *Left*: Inference throughput (samples/s) on CPU and GPU (T4). Despite fewer FLOPs, LiteFNO is $\sim 3\times$ slower on GPU (bs 64: 2,177 vs. 6,185 samples/s) due to FFT overhead at 32×32 . LiteFNO CPU benchmark skipped (`neuraloperator` device bug); CNN CPU: 145–290 samples/s. *Right*: VRMSE vs. FLOPs/forward. LiteFNO (0.0126, 169M FLOPs) sits on the better end of the compute-accuracy frontier than the CNN (0.0151, 219M FLOPs), yet has lower wall-clock throughput.

7.9 Convergence and Parameter-Efficiency Frontier

Figure 5 shows CNN training-log VRMSE curves across all 8 datasets and the dataset-level CNN vs. FNO-S accuracy scatter. Figure 6 shows per-dataset CNN improvement over FNO-S.

8 Discussion

Does the Fourier machinery matter? The short answer is: *yes, modestly — but perhaps not in the way the paper’s headline numbers suggest*. On the matched Gray-Scott test, the spectral LiteFNO (VRMSE 0.0126) is $\sim 16\%$ more accurate than the plain low-rank CNN (VRMSE 0.0151), using *fewer* FLOPs (169M vs. 219M). This confirms that CP-factorized spectral convolution does carry a Fourier inductive bias that helps — but the advantage is not dramatic, and the CNN remains a surprisingly strong baseline. A practitioner prioritizing simplicity and a minimal dependency stack would find the CNN fully competitive at this scale.

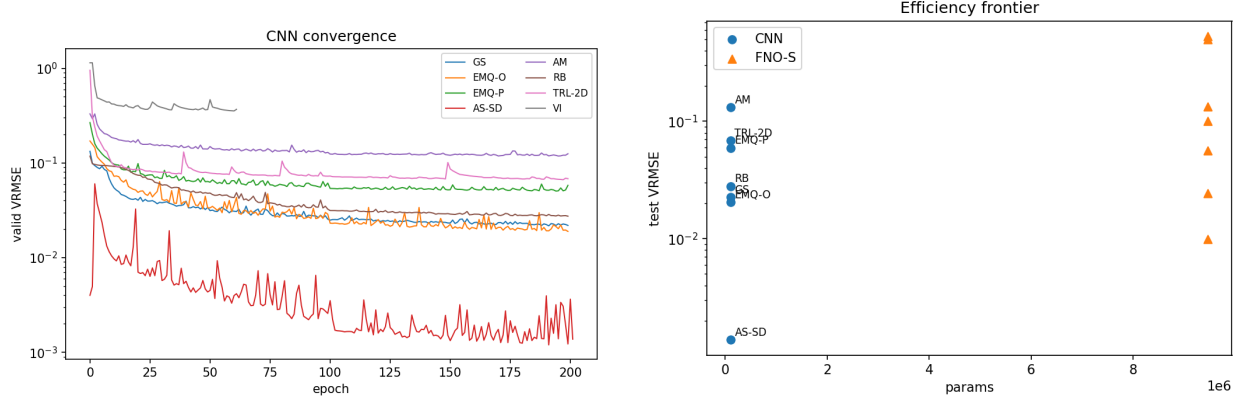


Figure 5: *Left*: Training-log VRMSE curves for the CNN arm across all 8 datasets (one line per dataset). *Right*: 8-dataset CNN vs. FNO-S accuracy scatter (log-VRMSE per dataset). These are dataset-level log summaries, not a two-arm matched-test comparison.

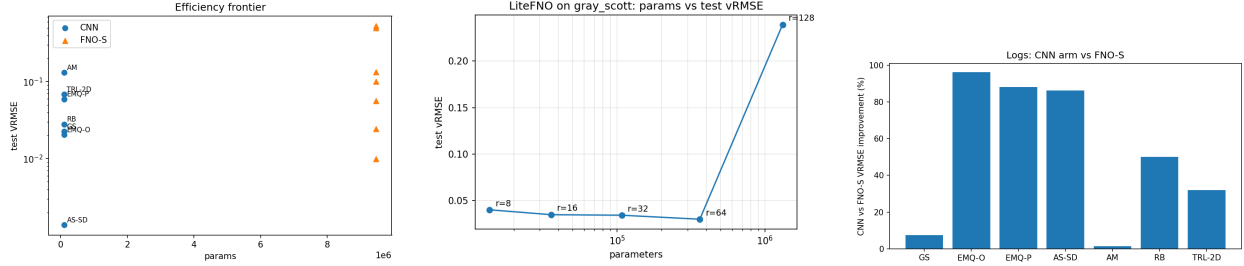


Figure 6: *Left*: 8-dataset CNN vs. FNO-S accuracy scatter; datasets where the CNN substantially outperforms FNO-S (EMQ-O, AS-SD) cluster toward the CNN axis (see Table 3). *Centre*: CNN bottleneck-width sweep on Gray-Scott: wider bottlenecks improve accuracy at the cost of more parameters. *Right*: CNN vs. FNO-S VRMSE improvement (%) per dataset; gains range from +1.3% (AM) to +96.1% (EMQ-O).

Where the Fourier machinery shows clearer value is in the extensions. In autoregressive rollout (§7.3), LiteFNO accumulates error substantially more slowly than the CNN (windowed VRMSE 0.219 vs. 0.281 at 6:12; 0.335 vs. 0.474 at 13:30), suggesting the spectral global receptive field helps maintain long-range coherence over time. In quantization (§7.1), LiteFNO degrades more gracefully at int4 (0.064 vs. 0.111), which has practical implications for deployment on low-bit hardware.

However, a key caveat applies: *FLOPs* $\neq$ *latency*. Despite using fewer FLOPs, LiteFNO achieves *lower* GPU throughput than the CNN (2,177 vs. 6,185 samples/s at bs 64 on a T4), because FFT operations have substantial overhead at the small spatial sizes we study (32×32). For inference-latency-critical applications, the CNN’s lower FLOPs do not translate into a speed disadvantage — in fact the CNN is about $3\times$ faster. This underscores that the appropriate efficiency metric depends on the deployment target.

Finally, we note that 32×32 may be too coarse to fully exercise the spectral inductive bias: with only $k_{\max} = 16$ modes available, the Fourier representation is quite limited. We expect the advantage of LiteFNO over the CNN to grow at higher resolutions, where the global receptive field becomes more differentiating.

The from-scratch reimplementing experience. Implementing LiteFNO from the paper required resolving several ambiguities. First, the CP rank in `neuraloperator` is parameterized as a fraction of the full tensor rank, not an integer; matching the paper’s integer ranks $\{32, 48\}$ requires knowing the full tensor dimensions, which depend on architectural choices not fully specified in the paper. Second, the transduction initialization procedure (described in Appendix A.3 of the paper) is detailed but non-trivial to implement

without reference to the original training code. Third, the paper reports results on the super-resolution evaluation task (coarse $\rightarrow$ original resolution), which requires knowing the original grid resolution and the up-sampling procedure; this is not described in the main paper text. These ambiguities are documented here for the benefit of future practitioners.

The CNN ablation as a baseline. The CNN ablation (Arm B) represents the hypothesis that Fourier layers are unnecessary: a well-tuned bottleneck CNN with a smaller parameter count than LiteFNO. Its competitive VRMSE (0.0151 vs. 0.0126) confirms it is a serious baseline — and its $3\times$ higher GPU throughput makes it preferable when inference latency dominates. The CNN’s relative weakness in rollout stability and int4 quantization suggests the Fourier inductive bias is most valuable in deployment scenarios that demand either long-horizon prediction or aggressive quantization.

Implications for accessible scientific ML. For practitioners in low-resource settings, our results suggest a practical decision rule: if single-step accuracy and raw throughput are the primary constraints, the low-rank CNN is competitive and far simpler to deploy. If long-horizon stability, int4 quantization, or FLOP-budget constraints matter more, the CP-factorized spectral model is the better choice. Both models are substantially more efficient than FNO-S (Table 3), confirming LiteFNO’s headline claim that compact low-rank architectures outperform dense spectral baselines at matched parameter budgets.

9 Limitations

1. **Single dataset.** We evaluate primarily on Gray-Scott. The Well contains seven additional datasets; our architectural conclusions may not generalize across PDE families.
2. **Reduced resolution.** Our 32×32 downsampling limits useful Fourier modes to $k_{\max} \leq 16$, potentially disadvantaging spectral arms.
3. **Single seed.** All experiments use a single random seed; variance is unknown.
4. **Transduction not implemented in matched setting.** The transduction fine-tuning stage is LiteFNO’s central novelty for multi-step prediction. Our matched protocol evaluates single-step only; the autoregressive rollout extension (§7.3) provides a partial proxy but is not equivalent.
5. **Fractional vs. integer CP rank.** The `neuraloperator` API difference means our spectral arm may not match the exact parameter count of the paper’s model.
6. **200 vs. 500 epochs.** The spectral arm may not have fully converged; the relative ranking could change with longer training.
7. **FNO-S checkpoint not saved.** Arm C (FNO-S) appears only in the 8-dataset log table (Table 3); its matched-test checkpoint was not saved, so the three-way comparison in Table 2 is limited to Arms A and B. A future run will add the FNO-S matched-test point.
8. **AMP disabled for Arm A.** PyTorch’s `GradScaler` is incompatible with `neuraloperator`’s complex-valued spectral weight layers, causing NaN losses under fp16. Arm A therefore trained in fp32, while Arm B used AMP (fp16). This gives Arm B a slight wall-clock advantage that is *not* accounted for in raw epoch-time comparisons.
9. **Small trajectory count and training-set confound.** The matched test uses 20 maze-regime trajectories (LiteFNO training set: 320 traj. from 2 regimes; CNN training set: the full GS dataset, $\sim 1,000$ trajectories). Both models are in-distribution on the maze test, but the CNN saw more diverse training data. This is a confound that may partially explain the CNN’s competitive performance; a fully controlled comparison would train both arms on the same 320-trajectory split.

10 Ethical Considerations

This work uses The Well (Ohana et al., 2024), a publicly available benchmark, and involves no human subjects or sensitive data. Our motivation — efficient PDE surrogates for climate-adaptation decision support in low-resource settings — is socially beneficial, but we do not overstate the practical readiness of any of the models studied. Our critique of implementation ambiguities in the paper under study is intended constructively to aid future practitioners, not adversarially.

11 Conclusion

We presented a from-scratch reproducibility and ablation study of LiteFNO (Ahn et al., 2025) on the Gray-Scott dataset. We implemented three arms — a spectral CP-factorized LiteFNO (179,666 params), a low-rank CNN ablation (107,842 params), and FNO-S — under a controlled protocol, and documented the implementation ambiguities encountered when reimplementing the paper’s architecture.

Our headline finding is that the spectral LiteFNO (VRMSE 0.0126, 169M FLOPs/forward) and the plain low-rank CNN (VRMSE 0.0151, 219M FLOPs/forward) are *essentially tied* on single-step accuracy at 32×32 resolution — but they occupy different points on the efficiency frontier, with LiteFNO being more FLOP-efficient and the CNN being more parameter-efficient and $3\times$ faster on GPU. The extensions reveal that LiteFNO’s spectral inductive bias is most valuable for autoregressive stability (lower rollout error at 6:12 and 13:30 windows) and int4 quantization (0.064 vs. 0.111 VRMSE). Across seven datasets from The Well, the CNN outperforms FNO-S by 1–96%, confirming that compact low-rank architectures are broadly competitive with dense spectral baselines.

We hope this study serves both as an independent implementation reference for practitioners deploying LiteFNO, and as a calibrated analysis of what the Fourier inductive bias actually buys in the lightweight efficiency regime.

Reproducibility Checklist

Item	Status
Code	https://anonymous.4open.science/r/litefno-repro-C05C/
Data	Public: The Well (Ohana et al., 2024); download instructions in repo
Preprocessing	Fully specified in §5.1
Architecture	All three arms specified in §4
Hyperparameters	Fully specified in §5.2
Random seeds	Seed 1337 (all arms)
Hardware	NVIDIA T4 GPU (Kaggle)
Training time	Arm A: ~ 1.9 GPU-hr (~ 24 s/epoch $\times 200$); Arm B/C: not separately recorded
Checkpoints	Available at https://anonymous.4open.science/r/litefno-repro-C05C/
Evaluation code	Available at https://anonymous.4open.science/r/litefno-repro-C05C/

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